

UNIT 11

TRIGONOMETRY



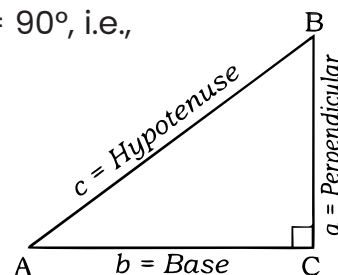
The word trigonometry is a Greek word. It is an important branch of mathematics that deals with the relations of measures of sides and angles of a triangle. There are six elements of triangle. i.e. three sides and three angles. If the measures of any three elements with at least one side are known, measure of the other elements can be found.

TRIGONOMETRIC RATIOS OF ACUTE ANGLES

Consider a right-angled triangle ACB in which $m\angle C = 90^\circ$, i.e., a right angle.

Measures of the sides opposite to $\angle A$, $\angle B$, and $\angle C$ are represented by a , b , and c respectively, i.e., $m\overline{BC} = a$, $m\overline{CA} = b$, and $m\overline{AB} = c$.

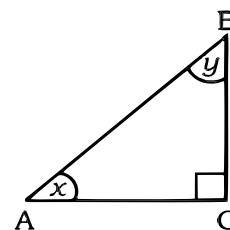
In a right triangle, the ratio of any of its two sides is called a **trigonometric ratio**.



- (i) Sine of $m\angle A = \sin(m\angle A) = \frac{\text{Opp. Side of } m\angle A}{\text{Hypotenuse}} = \frac{m\overline{BC}}{m\overline{AB}} = \frac{a}{c}$
- (ii) Cosine of $m\angle A = \cos(m\angle A) = \frac{\text{Adj. Side of } m\angle A}{\text{Hypotenuse}} = \frac{m\overline{AC}}{m\overline{AB}} = \frac{b}{c}$
- (iii) Tangent of $m\angle A = \tan(m\angle A) = \frac{\text{Opp. Side of } m\angle A}{\text{Adjacent Side of } m\angle A} = \frac{m\overline{BC}}{m\overline{AC}} = \frac{a}{b}$
- (iv) Cotangent of $m\angle A = \cot(m\angle A) = \frac{\text{Adjacent Side of } m\angle A}{\text{Opposite Side of } m\angle A} = \frac{m\overline{AC}}{m\overline{BC}} = \frac{b}{a}$
- (v) Secant of $m\angle A = \sec(m\angle A) = \frac{\text{Hypotenuse}}{\text{Adjacent Side of } m\angle A} = \frac{m\overline{AB}}{m\overline{AC}} = \frac{c}{b}$
- (vi) Cosecant of $m\angle A = \operatorname{cosec} m\angle A = \frac{\text{Hypotenuse}}{\text{Opposite Side of } m\angle A} = \frac{m\overline{AB}}{m\overline{BC}} = \frac{c}{a}$

By considering the above ratios, it is observed that:

- (i) $\sin m\angle A = \frac{1}{\operatorname{cosec} m\angle A}$ and $\operatorname{cosec} m\angle B = \frac{1}{\sin m\angle B}$
- (ii) $\cos m\angle A = \frac{1}{\sec m\angle A}$ and $\sec m\angle B = \frac{1}{\cos m\angle B}$
- (iii) $\tan m\angle A = \frac{1}{\cot m\angle A}$ and $\cot m\angle B = \frac{1}{\tan m\angle B}$



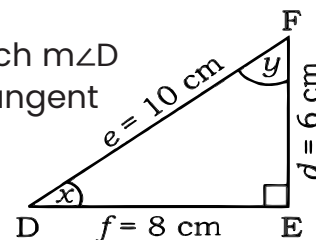
It is also observed that:

$$(a) \frac{\sin m\angle A}{\cos m\angle A} = \tan m\angle A \quad (b) \frac{\cos m\angle B}{\sin m\angle B} = \cot m\angle B$$

$$\text{or} \quad \frac{\sin x}{\cos x} = \tan x \quad \text{or} \quad \frac{\cos y}{\sin y} = \cot y$$

Example 1. Consider DEF, a right-angled triangle in which $m\angle D = x$ and $m\angle F = y$. Find the values of sine, cosine, and tangent of angle D.

Solution:



$$(i) \sin x = \frac{\text{Opposite Side of } m\angle D}{\text{Hypotenuse}} = \frac{m\overline{EF}}{m\overline{DF}} = \frac{d}{e} = \frac{6}{10} = \frac{3}{5}$$

$$(ii) \cos x = \frac{\text{Adjacent Side of } m\angle D}{\text{Hypotenuse}} = \frac{m\overline{DE}}{m\overline{DF}} = \frac{f}{e} = \frac{8}{10} = \frac{4}{5}$$

$$(iii) \tan x = \frac{\text{Opposite Side of } m\angle D}{\text{Adjacent Side}} = \frac{m\overline{EF}}{m\overline{DE}} = \frac{d}{f} = \frac{6}{8} = \frac{3}{4}$$

Example 2. In the given $\triangle ABC$, verify the following relations between trigonometric ratios.

$$(i) \frac{\cos x}{\sin x} = \cot x \quad (ii) \frac{\sin y}{\cos y} = \tan y \quad (iii) \tan x \cot x = 1$$

Solution:

$$(i) \cos x / \sin x = \cot x$$

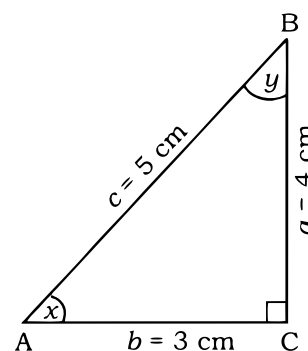
Here,

$$\text{L.H.S} = \frac{\cos x}{\sin x} = \frac{b/c}{a/c} = \frac{3/5}{4/5} = \frac{3}{5} \times \frac{5}{4} = \frac{3}{4}$$

$$\text{R.H.S} = \cot x = \frac{\text{Adjacent Side of } m\angle A}{\text{Opposite Side of } m\angle A} = \frac{m\overline{AB}}{m\overline{BC}} = \frac{b}{a} = \frac{3}{4}$$

Hence

$$\text{L.H.S} = \text{R.H.S}$$



$$(ii) \sin y / \cos y = \tan y$$

Here,

$$\text{L.H.S} = \frac{\sin y}{\cos y} = \frac{\text{Opposite Side of } m\angle B}{\text{Adjacent Side of } m\angle B} = \frac{b/c}{a/c} = \frac{3/5}{4/5} = \frac{3}{5} \times \frac{5}{4} = \frac{3}{4}$$

$$\text{R.H.S} = \tan y = \frac{\text{Opposite Side of } m\angle B}{\text{Adjacent Side of } m\angle B} = \frac{m\overline{AC}}{m\overline{BC}} = \frac{b}{a} = \frac{3}{4}$$

Hence

$$\text{L.H.S} = \text{R.H.S}$$

$$(iii) \tan x \cot x = 1$$

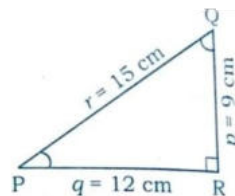
$$\begin{aligned} \text{L.H.S} = \tan x \cot x &= \frac{\text{Opposite Side of } m\angle A}{\text{Adjacent Side of } m\angle A} \times \frac{\text{Adjacent Side of } m\angle A}{\text{Opposite Side of } m\angle A} \\ &= \frac{a}{b} \times \frac{b}{a} = \frac{4}{3} \times \frac{3}{4} = \frac{1}{1} = 1 \end{aligned}$$

Hence

$$\text{L.H.S} = \text{R.H.S}$$

EXERCISE 11.1

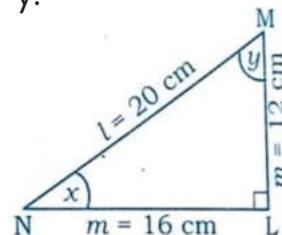
1. In the given right-angled triangle PQR, find all the trigonometric ratios for both the acute angles.



2. In the given right-angled triangle LMN, $m\angle M = x$, $m\angle N = y$. Find the values of the following trigonometric ratios:

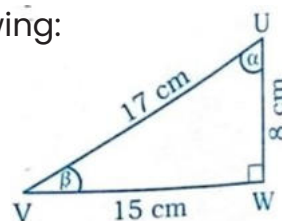
- | | | |
|-------------------------------|-------------------------------|----------------|
| (i) $\sec x$ | (ii) $\cos y$ | (iii) $\tan x$ |
| (iv) $\operatorname{cosec} y$ | (v) $\sec x$ | (vi) $\cot y$ |
| (vii) $\cos x$ | (viii) $\sin y$ | (ix) $\cot x$ |
| (x) $\sec y$ | (xi) $\operatorname{cosec} x$ | (xii) $\tan y$ |

Which of the above trigonometric ratios are equal?



3. In the given right-angled triangle UVW, verify the following:

- | | |
|---|--|
| (i) $\sin \alpha \cdot \operatorname{cosec} \alpha = 1$ | (ii) $\sec \beta \cdot \cos \beta = 1$ |
| (iii) $\tan \alpha \cdot \cot \alpha = 1$ | (iv) $\sin \beta / \cos \beta = \tan \beta$ |
| (v) $\cos \alpha / \sin \alpha = \cot \alpha$ | (vi) $\cos \alpha \cdot \operatorname{cosec} \alpha = \cot \alpha$ |

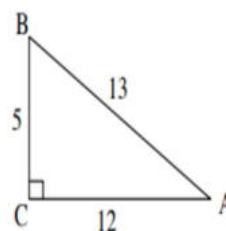


Q. Find each of the following ratios for the triangles shown below.

(a) $\sin A =$ _____ (b) $\tan B =$ _____

(c) $\cos A =$ _____ (d) $\tan A =$ _____

(e) $\cos B =$ _____ (f) $\sin B =$ _____



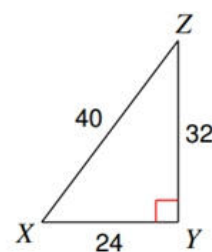
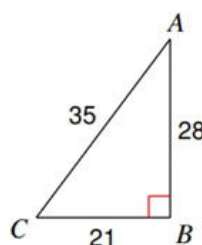
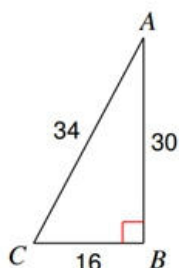
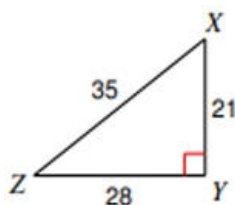
Q. Find the value of each trigonometric ratio.

1) $\tan Z$

2) $\cos C$

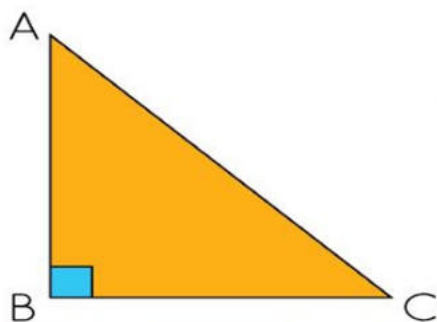
3) $\sin C$

4) $\tan X$



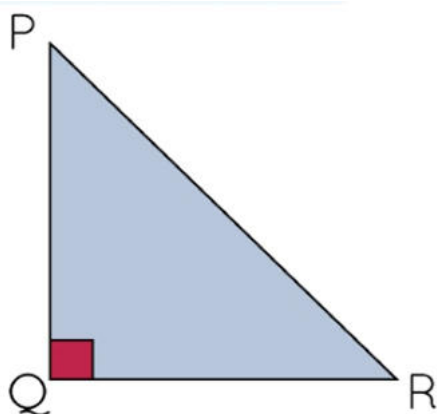
1. In a triangle ABC, right-angled at B, AC = 5 inches and angle A = 60° .

Find: Sin A



2. In a triangle PQR, right-angled at Q, PR = 8 inches, and PQ = 4 inches.

Find: angle P, angle R

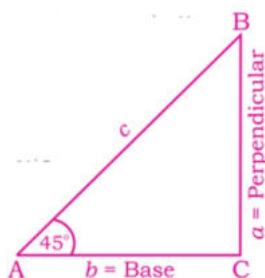


TRIGONOMETRIC RATIOS OF ACUTE ANGLES

Trigonometric Ratios of an angle of measure of 45°

Consider the right-angled triangle ABC with right angle at point C and $m\angle A = 45^\circ$.

So, $m\angle B = 90^\circ - m\angle A = 90^\circ - 45^\circ = 45^\circ$.



$m\angle A = m\angle B = 45^\circ$, then $m\angle C = 90^\circ$

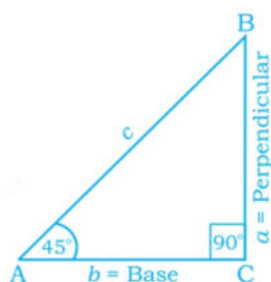
We know that: $(\text{Hyp})^2 = (\text{Perp})^2 + (\text{Base})^2$

or $a = b$ (Pythagoras Theorem)

or $c^2 = a^2 + b^2$

or $c^2 = b^2 + b^2 = 2b^2$ ($\because a = b$)

$(c)^2 = (\sqrt{2} b)^2$, i.e., $c = \sqrt{2} b$.



$$(i) \sin 45^\circ = \frac{b}{\sqrt{2}b} = \frac{1}{\sqrt{2}}$$

$$(v) \sec 45^\circ = \frac{\sqrt{2}b}{b} = \sqrt{2}$$

$$(ii) \cos 45^\circ = \frac{b}{c} = \frac{b}{\sqrt{2}b} = \frac{1}{\sqrt{2}}$$

$$(vi) \operatorname{cosec} 45^\circ = \frac{\sqrt{2}b}{b} = \sqrt{2}$$

$$(iii) \tan 45^\circ = \frac{a}{b} = 1 \text{ or } \tan 45^\circ = 1$$

$$(iv) \cot 45^\circ = \frac{b}{a} = 1 \text{ or } \cot 45^\circ = 1$$

Trigonometric Ratios of an acute angle measuring 30°

Consider the right-angled triangle ABC, with right angle at point C and $m\angle A = 30^\circ$.

So, $m\angle B = 90^\circ - m\angle A = 90^\circ - 30^\circ = 60^\circ$.

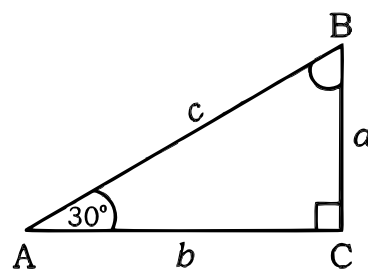
Since $c = 2a$ in such a triangle,

By Pythagoras Theorem, $c^2 = a^2 + b^2$

Putting $c = 2a$, we get: $(2a)^2 = a^2 + b^2$

or $4a^2 - a^2 = b^2$ or $3a^2 = b^2$

So, $b = \sqrt{3}a$



$$(i) \sin 30^\circ = \frac{\text{Opposite Side of } m\angle A}{\text{Hypotenuse}} = \frac{m \overline{BC}}{m \overline{AB}} = \frac{a}{c} = \frac{a}{2a} = \frac{1}{2}$$

$$(ii) \cos 30^\circ = \frac{\text{Adjacent Side of } m\angle A}{\text{Hypotenuse}} = \frac{m \overline{AC}}{m \overline{AB}} = \frac{b}{c} = \frac{\sqrt{3}a}{2a} = \frac{\sqrt{3}}{2}$$

$$(iii) \tan 30^\circ = \frac{\text{Opposite Side of } m\angle A}{\text{Adjacent Side of } m\angle A} = \frac{m \overline{BC}}{m \overline{AC}} = \frac{a}{b} = \frac{a}{\sqrt{3}a} = \frac{1}{\sqrt{3}}$$

$$(iv) \cot 30^\circ = \frac{\text{Adjacent Side of } m\angle A}{\text{Opposite Side of } m\angle A} = \frac{m \overline{AC}}{m \overline{BC}} = \frac{b}{a} = \frac{\sqrt{3}a}{a} = \sqrt{3}$$

$$(v) \sec 30^\circ = \frac{\text{Hypotenuse}}{\text{Adjacent Side of } m\angle A} = \frac{m \overline{AB}}{m \overline{AC}} = \frac{c}{b} = \frac{2a}{\sqrt{3}a} = \frac{2}{\sqrt{3}}$$

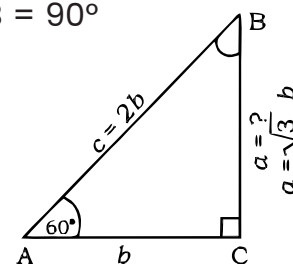
$$(vi) \operatorname{cosec} 30^\circ = \frac{\text{Hypotenuse}}{\text{Opposite Side of } m\angle A} = \frac{m \overline{AB}}{m \overline{BC}} = \frac{c}{a} = \frac{2a}{a} = 2$$

Trigonometric Ratios of an acute angle measuring 60°

Consider the right-angled triangle ABC, with right angle at point C and $m\angle A = 60^\circ$. In the given right-angled triangle ABC, $m\angle B = 90^\circ$

- $m\angle A = 90^\circ - 60^\circ = 30^\circ$.

Thus, $m\angle A = 60^\circ$, $m\angle B = 30^\circ$, and $m\angle C = 90^\circ$.



(By Pythagoras Theorem)

$$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$$

$$c^2 = b^2 + a^2$$

$$\text{or } (2b)^2 = b^2 + a^2 \text{ (putting } c = 2b)$$

$$\text{or } 4b^2 - b^2 = a^2$$

$$\Rightarrow a^2 = 3b^2 \quad \text{or} \quad \sqrt{a^2} = \sqrt{3b^2}$$

$$\text{or } a = \sqrt{3} b$$

$$(i) \sin m\angle A = \sin 60^\circ = \frac{\text{Opposite Side of } m\angle A}{\text{Hypotenuse}} = \frac{a}{c} = \frac{\sqrt{3} b}{2b} = \frac{\sqrt{3}}{2}$$

$$(ii) \cos m\angle A = \cos 60^\circ = \frac{\text{Adjacent Side of } m\angle A}{\text{Hypotenuse}} = \frac{b}{2b} = \frac{1}{2}$$

$$(iii) \tan m\angle A = \tan 60^\circ = \frac{\text{Opposite Side of } m\angle A}{\text{Adjacent Side of } m\angle A} = \frac{a}{b} = \frac{\sqrt{3} b}{b} = \sqrt{3}$$

$$(iv) \cot m\angle A = \cot 60^\circ = \frac{\text{Adjacent Side of } m\angle A}{\text{Opposite Side of } m\angle A} = \frac{b}{a} = \frac{b}{\sqrt{3} b} = \frac{1}{\sqrt{3}}$$

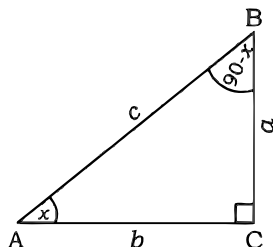
$$(v) \sec m\angle A = \sec 60^\circ = \frac{\text{Hypotenuse}}{\text{Adjacent Side of } m\angle A} = \frac{c}{b} = \frac{2b}{b} = 2$$

$$(vi) \operatorname{cosec} m\angle A = \operatorname{cosec} 60^\circ = \frac{\text{Hypotenuse}}{\text{Opp. Side of } m\angle A} = \frac{c}{a} = \frac{2b}{\sqrt{3} b} = \frac{2}{\sqrt{3}}$$

Angle θ	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\cot \theta$	$\sec \theta$	$\operatorname{cosec} \theta$
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$	$\sqrt{3}$	$\frac{2}{\sqrt{3}}$	2
45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1	1	$\sqrt{2}$	$\sqrt{2}$
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{1}{\sqrt{3}}$	2	$\frac{2}{\sqrt{3}}$

TRIGONOMETRIC RATIOS OF COMPLEMENTARY ANGLES

We have already learnt that the acute angles of a right-angled triangle are complementary angles. ABC is a right-angled triangle, with $m\angle C = 90^\circ$, $m\angle A + m\angle B = 90^\circ$. Let $m\angle A = x$, then $m\angle B = 90^\circ - m\angle A$ or $m\angle B = 90^\circ - x$.



From the above figure, we see that:

$$(i) \sin x = \frac{a}{c} = \cos (90 - x) \Rightarrow \sin x = \cos (90 - x)$$

$$(ii) \cos x = \frac{b}{c} = \sin (90 - x) \Rightarrow \cos x = \sin (90 - x)$$

Also from the above table we see that:

$$\sin 30^\circ = \frac{1}{2} = \cos (90 - 30)^\circ \Rightarrow \sin 30^\circ = \cos 60^\circ$$

$$\text{And } \cos 30^\circ = \frac{\sqrt{3}}{2} = \sin (90 - 30)^\circ \Rightarrow \cos 30^\circ = \sin 60^\circ$$

From the above discussion, we conclude that:

- (a) Sin and Cos of complementary angles are equal.
- (b) Tan and Cot of complementary angles are equal.
- (c) Sec and Cosec of complementary angles are equal.

Example. Find the value of the following:

- (i) $\sin 30^\circ \times \cos 45^\circ + \cos 60^\circ \times \sin 45^\circ$
- (ii) $\tan 30^\circ \times \cot 60^\circ + \cot 30^\circ \times \tan 60^\circ$
- (iii) $\sec 30^\circ \times \operatorname{cosec} 45^\circ + \sec 45^\circ \times \operatorname{cosec} 60^\circ$

Solution:

$$(i) \sin 30^\circ \times \cos 45^\circ + \cos 60^\circ \times \sin 45^\circ$$

Putting the values from the table in the expression given above; we get $\sin 30^\circ \times \cos 45^\circ + \cos 60^\circ \times \sin 45^\circ$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \times \frac{1}{\sqrt{2}} \\ = \frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{1+1}{2\sqrt{2}} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$(ii) \tan 30^\circ \times \cot 60^\circ + \cot 30^\circ \times \tan 60^\circ$$

$\tan 30^\circ = 1/\sqrt{3}$, $\tan 60^\circ = \sqrt{3}$, $\cot 30^\circ = \sqrt{3}$
and $\cot 60^\circ = 1/\sqrt{3}$ in the given expression,

$$\tan 30^\circ \times \cot 60^\circ + \cot 30^\circ \times \tan 60^\circ$$

$$= \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}} + \sqrt{3} \times \sqrt{3} = \frac{1}{\sqrt{3} \times 3} + \sqrt{3 \times 3}$$

$$= \frac{1}{3} + \frac{3}{1} = \frac{1+9}{3} = \frac{10}{3} = 3 \frac{1}{3} = 3.333$$

(iii) $\sec 30^\circ \times \operatorname{Cosec} 60^\circ + \sec 45^\circ \times \operatorname{Cosec} 45^\circ$

We know $\sec 30^\circ = 2 / \sqrt{3} = \operatorname{Cosec} 60^\circ$ and $\operatorname{Cosec} 45^\circ = \sqrt{2} = \sec 45^\circ$.

So, $\sec 30^\circ \times \operatorname{Cosec} 45^\circ + \sec 45^\circ \times \operatorname{Cosec} 60^\circ$

$$= \frac{2}{\sqrt{3}} \times \frac{2}{\sqrt{3}} + \frac{\sqrt{2} \times \sqrt{2}}{1} = \frac{2 \times 2}{\sqrt{3} \times 3} + \frac{\sqrt{2 \times 2}}{1}$$

$$= \frac{4}{3} + \frac{2}{1} = \frac{4+6}{3} = \frac{10}{3} = 3 \frac{1}{3} = 3.333$$

Example 2. Find:

(a) Value of x when the value of $\sin x$ is $1 / 2$

(b) Value of y when the value of $\tan y = 1$

Solution:

(a) We know that: $\sin 30^\circ = 1 / 2$ (from the table)

But here $\sin x = 1 / 2$

This implies that: $\sin x = \sin 30^\circ \Rightarrow x = 30^\circ$.

(b) We know that, $\tan 45^\circ = 1$ (from the table)

But here $\tan y = 1 \Rightarrow \tan y = \tan 45^\circ \Rightarrow y = 45^\circ$.

EXERCISE 11.2

1. Find the value of x , when the value of $\cos x$ is:

(i) $\sqrt{3} / 2$

(ii) $1 / \sqrt{2}$

(iii) $1 / 2$

2. Find the value of y , when the value of $\tan y$ is:

(i) $\sqrt{3}$

(ii) 1

(iii) $1 / \sqrt{3}$

3. Find the value of $m\angle A$, when the value of $\operatorname{Cosec} m\angle A$ is:

(i) $2 / 1$

(ii) $\sqrt{2}$

(iii) $2 / \sqrt{3}$

4. Find the value of the following:

(i) $\sin 30^\circ \times \sin 60^\circ + \cos 30^\circ \times \cos 60^\circ$

(ii) $\sec 30^\circ \times \operatorname{Cosec} 60^\circ + \tan 60^\circ \times \cot 30^\circ$

(iii) $\cos 30^\circ \times \sin 30^\circ + \operatorname{Cosec} 30^\circ \times \sec 30^\circ$

(iv) $\sin 60^\circ \times \sin 45^\circ - \cos 30^\circ \times \cos 45^\circ$

(v) $\tan 30^\circ \times \cot 60^\circ - \sin 45^\circ \times \cos 45^\circ$

(vi) $\sec 45^\circ \times \operatorname{Cosec} 45^\circ + \sec 30^\circ \times \operatorname{Cosec} 60^\circ$

(vii) $\cot 60^\circ \times \sec 30^\circ - \operatorname{Cosec} 60^\circ \times \tan 30^\circ$

Determine the exact values of the following

- (a) $\sec^2 45^\circ \csc^2 45^\circ - 1$ (b) $\sin 30^\circ \tan 45^\circ + \tan 30^\circ \sin 60^\circ$
 (c) $\cos 30^\circ \sin 45^\circ + \sin 30^\circ \tan 30^\circ$ (d) $2 \sin 30^\circ \cos 30^\circ$
 (e) $\sin 30^\circ \cos 30^\circ + \sin 60^\circ \cos 60^\circ$ (f) $\tan 45^\circ - (\sin 45^\circ / \cos 45^\circ)$
 (g) $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$ (h) $\sin 30^\circ + \cos 60^\circ + \sec 60^\circ$

RIGHT ANGLED TRIANGLES

Solve Right Angled Triangles using Trigonometric Ratios

The process of finding the unknown elements of a triangle is known as solving the triangle.

Case I: Measure of one side and one acute angle is given.

Example 1. Solve right-angled triangle ABC, with $m\angle C = 90^\circ$, $m\angle A = 60^\circ$ and the measure of its side opposite to $m\angle A = 12$ cm.

Solution: Here $m\angle C = 90^\circ$, $m\angle A = 60^\circ$ and $a = 12$ cm.

Now we have to find $m\angle B$, $m AC = b$ and $m AB = c$

We also know that in a right-angled triangle, acute angles are complementary.

Therefore, $m\angle A + m\angle B = 90^\circ$

$m\angle B = 90^\circ - 60^\circ = 30^\circ$

$$\tan m\angle A = \tan 60^\circ = \frac{\text{Opposite Side of } m\angle A}{\text{Adjacent Side of } m\angle A} = \frac{a}{b} = \frac{12}{b}$$

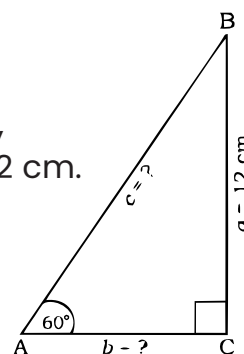
$$\text{or } \sqrt{3} = \frac{12}{b} \text{ or } b = \frac{12}{\sqrt{3}} = 4\sqrt{3} = 4 \times 1.732 = 6.928 \text{ cm}$$

$$\text{Again, } \sin m\angle A = \sin 60^\circ = \frac{a}{c} = \frac{12}{c}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{12}{c} \Rightarrow c\sqrt{3} = 12 \times 2$$

$$\Rightarrow c = \frac{24}{\sqrt{3}} = \frac{24\sqrt{3}}{\sqrt{3}\sqrt{3}} = \frac{24\sqrt{3}}{3} = 8\sqrt{3}$$

Thus, $m\angle B = 30^\circ$, $m AC = 4\sqrt{3} = 6.9$ cm, $m AB = 13.8$ cm



Case II: Measure of Hypotenuse and one acute angle is given.

Example 2. Solve a right triangle DEF, when $m\angle E = 90^\circ$, $m\angle D = 30^\circ$ and $m DF$ (Hypotenuse) = 10 cm.

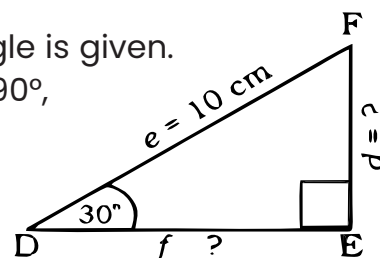
Solution: Here DEF is a right triangle, in which

$m\angle E = 90^\circ$, $m\angle D = 30^\circ$ and Hypotenuse = $m DF = 10$ cm

We have to find $m\angle F$, $m DE = f$, $m EF = d$

We know that:

$$\sin m\angle D = \sin 30^\circ = \frac{\text{Opposite Side of } m\angle D}{\text{Hypotenuse}} = \frac{d}{e} = \frac{d}{10}$$



or $\sin 30^\circ = \frac{d}{10}$

or $\frac{1}{2} = \frac{d}{10}$, or $d = 5 \text{ cm}$

Also $\frac{f}{e} = \cos m\angle D \Rightarrow \frac{f}{e} = \cos 30^\circ$

$\Rightarrow \frac{f}{10} = \frac{\sqrt{3}}{2} \Rightarrow f = \frac{10\sqrt{3}}{2} = 5\sqrt{3} = 8.7 \text{ cm}$

$m\angle D + m\angle F = 90^\circ$ (Complementary angles)

So,

$m\angle F = 90^\circ - m\angle D = 90^\circ - 30^\circ = 60^\circ$

Hence, the solution of the given $\triangle DEF$ is as under:

$m\angle F = 60^\circ$, $m\angle D = 30^\circ$, $m\angle E = 90^\circ$, $DE = 5\sqrt{3} = 8.7 \text{ cm}$ and, $EF = 5 \text{ cm}$

Case III: The measure of two sides is given.

Example 3. Solve a right-angled triangle PQR with $m\angle Q = 90^\circ$ and measure of its perpendicular is $4\sqrt{3} \text{ cm}$ and of its base is 4 cm .

Solution: Here $\triangle PQR$ is a right-angled triangle, in which $m\angle Q = 90^\circ$, $m\angle P = 4 \text{ cm}$, $m\angle R = 4\sqrt{3} \text{ cm}$.

Now we have to find $m\angle P$, $m\angle R$ and $m\angle Q = \text{Hypotenuse}$.

According to Pythagoras Theorem:

$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$

$q^2 = r^2 + p^2$

$\Rightarrow (q)^2 = (4)^2 + (4\sqrt{3})^2 = 16 + 48 = 64$

$\therefore \text{Hypotenuse} = q = \sqrt{64} = 8 \text{ cm}$.

So, $m\angle P = 8 \text{ cm}$.

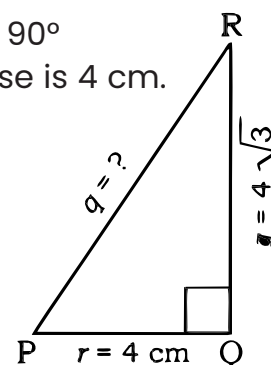
We know that $\tan m\angle P = \frac{\text{Opposite Side of } m\angle P}{\text{Adjacent Side}} = \frac{4\sqrt{3}}{4} = \sqrt{3}$.

$\therefore m\angle P = 60^\circ$

Then $m\angle R = 90^\circ - 60^\circ = 30^\circ$.

Hence solution of the given triangle is as under:

$m\angle P = 60^\circ$, $m\angle R = 30^\circ$ and $\text{Hyp} = 8 \text{ cm}$.



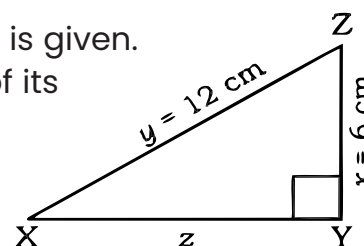
Case IV: The measures of one side and Hypotenuse is given.

Example 4. Solve $\triangle XYZ$ when $m\angle Y = 90^\circ$, measure of its Perpendicular is 6 cm and Hypotenuse is 12 cm .

Solution:

Here we have to find the unknown elements.

$m\angle X = ?$, $m\angle Z = ?$ and $m\angle Y = z = ?$



We know that $\sin m\angle X = \frac{\text{Opposite Side of } m\angle X}{\text{Hypotenuse}} = \frac{m\overline{YZ}}{m\overline{XZ}} = \frac{6}{12} = \frac{1}{2}$.

Thus, $\sin m\angle X = 1/2 \Rightarrow m\angle X = 30^\circ$.

Now $m\angle Z = 90^\circ - m\angle X = 90^\circ - 30^\circ = 60^\circ$.

Now:

$$z/y = \cos m\angle X$$

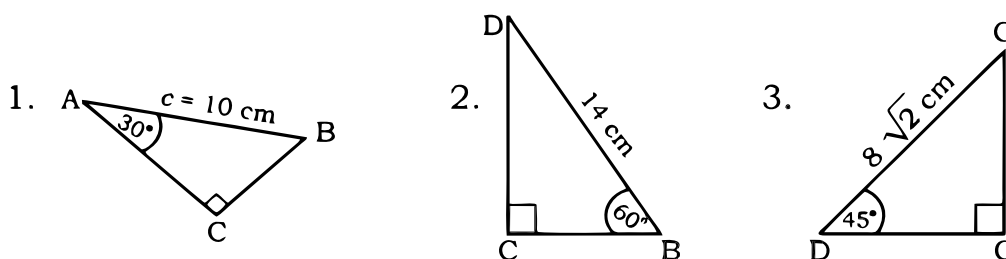
$$\Rightarrow z/12 = \cos 30^\circ = \sqrt{3}/2$$

$$\Rightarrow z = (\sqrt{3}/2) \times 12 = 6\sqrt{3}$$

Hence, $m\angle X = 30^\circ$, $m\angle Z = 60^\circ$ and $mXY = 6\sqrt{3}$ cm.

EXERCISE 11.3

A. Find unknown elements of the following right-angled triangles.

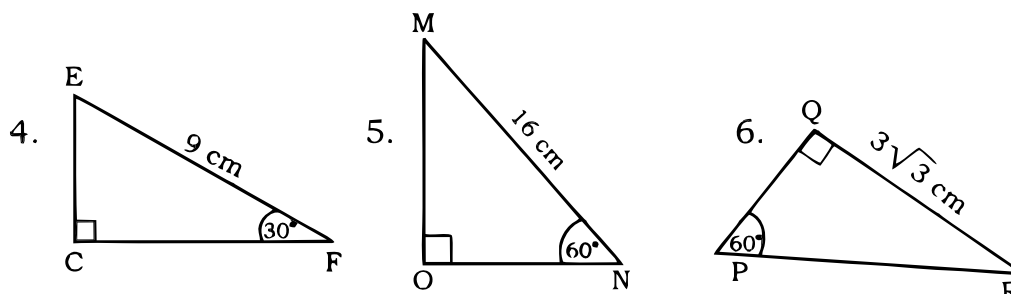


B. Solve the following right-angled triangles:

1. $\triangle ABC$, $m\angle C = 90^\circ$, $m\angle A = 30^\circ$ and Perpendicular $m\overline{BC} = 3$ cm

2. $\triangle DEF$, $m\angle E = 90^\circ$, $m\angle D = 60^\circ$ and Base $m\overline{DE} = 4$ cm

3. $\triangle LMN$, $m\angle M = 90^\circ$, $m\angle N = 45^\circ$ and Perpendicular $m\overline{ML} = 8$ cm



C. Solve the following right-angled triangles.

1. $\triangle LMN$, $m\angle M = 90^\circ$, perpendicular side $l = 4$ cm and base side $n = 4/\sqrt{3}$

2. $\triangle PQR$, $m\angle Q = 90^\circ$, perpendicular $p = 6/\sqrt{3}$ cm, and base side $r = 6$ cm

D. Solve the following right-angled triangles.

1. $\triangle ABC$, $m\angle C = 90^\circ$, $a = 5$ cm and Hypotenuse $c = 10$ cm

2. $\triangle DEF$, $m\angle C = 90^\circ$, $d = 7/\sqrt{3}$ cm and Hypotenuse $f = 14$ cm

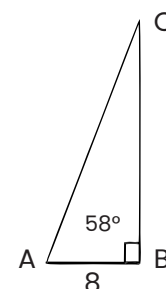
1. In right triangle ABC, $m\angle A = 58^\circ$ and $AB = 8$ cm.

Find the length of each of the following.

Round your answers to the nearest tenth.

(a) $BC =$ _____

(b) $AC =$ _____



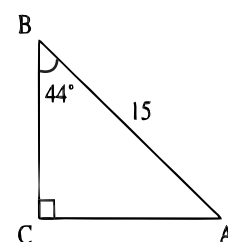
2. In right triangle ABC, $m\angle B = 44^\circ$ and $AB = 15$ cm.

Find the length of each of the following.

Round your answers to the nearest tenth.

(a) $AC =$ _____

(b) $BC =$ _____



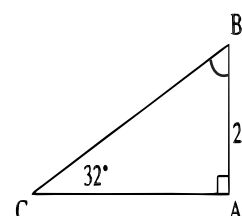
3. In right triangle ABC, $m\angle C = 32^\circ$ and $AB = 24$ cm.

Find the length of each of the following.

Round your answers to the nearest tenth.

(a) $AC =$ _____

(b) $BC =$ _____

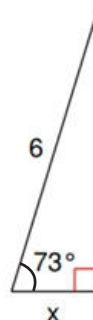


E. Find the missing side.

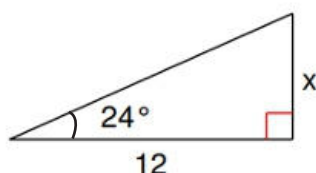
1)



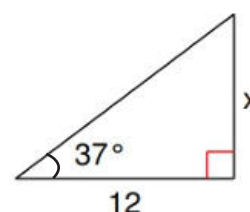
2)



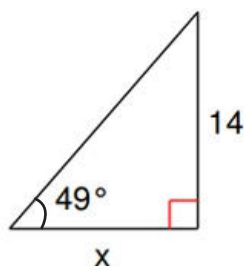
3)



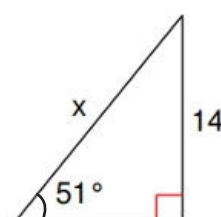
4)



5)

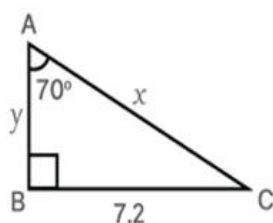


6)



F. Use trigonometric ratios to find the missing sides in the given right triangles

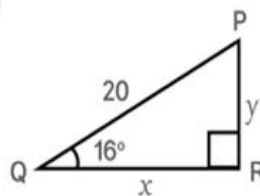
1



$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

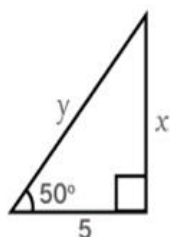
2



$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

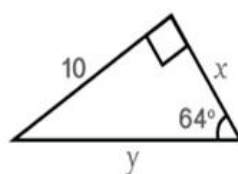
3



$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

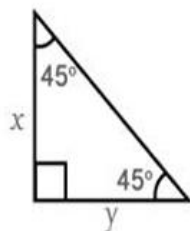
4



$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

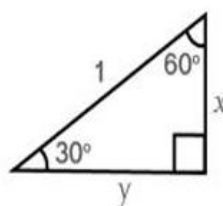
5



$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

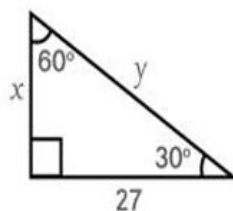
6



$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

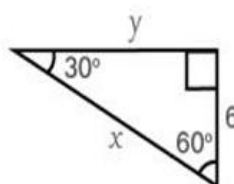
7



$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

8



$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$